Requirements Analysis

Purpose and Scope

This document records the assumptions the design is built on and the requirements derived from them. It is the first of two deliverables; the data model design is documented separately.

Scope. This project designs a consolidation database for [Client firm], a Lugano-based independent wealth manager, that ingests custody and portfolio data from its two principal banking relationships — Corner (delivered as Allocare GDF files) and Julius Baer (delivered via BJB Digital Data Services: a positions/transactions feed, camt.053 cash statements, and eDocs PDF packages) — and consolidates them into a single relational data model serving consolidated reporting and cross-source reconciliation. The database is a downstream layer: Allocare AMS remains the system of record for instrument and portfolio reference data, and the database does not write back to either bank. It is built around a source-aware staging area that lands each feed in its native shape, a canonical identity layer that maps both banks' instruments and accounts to common keys, and a conformed core (instruments, portfolios/accounts, transactions, positions, cash, and a document register) into which both sources are normalised. The design assumes direct ingestion of both sources over SFTP at small scale (low-hundreds of portfolios) and is structured so that the Julius Baer feed layout, the choice of platform, and the ingestion route can be finalised once the firm confirms them, without redesigning the core.

Conventions. Assumptions are identified **A1–A9**; requirements are identified **FR-*n*** (functional) and **NFR-*n*** (non-functional) for traceability, and cross-reference the assumptions they depend on.

1 Domain Assumptions

The design is currently built on the assumptions below. They state the *strong* version of each open question (direct ingestion of both sources, Allocare retained), because those drive the most capable design and degrade gracefully if the firm's answer turns out to be simpler.

A1 — Ingestion route — design for Route B (direct ingestion from both banks).

We assume the firm receives Corner data as Allocare GDF files *and* Julius Baer data directly via BJB DDS (CSV, camt.053, eDocs), loaded in parallel into the firm's own database. Route B is the supersetting case: it still works if BJB in fact flows through Allocare (the BJB parsers simply go unused), whereas a Route-A design would need rebuilding if BJB is ingested directly.

A2 — Allocare AMS remains the system of record for reference data.

We assume Allocare AMS stays in place and authoritative for instrument and portfolio standing data (P01, P03, P02/P46, P31/P32). The new database is a downstream consolidation and reconciliation layer, not a replacement — and is needed because BJB ships no security master.

A3 — Infrastructure — SFTP pull, scheduled batch, single relational database.

Absent specification, we assume SFTP delivery on a batch cycle aligned to BJB's pre-08:00 CET, T-1 window; a single RDBMS (MS SQL Server a reasonable default, since Allocare runs on it); and staging-then-core processing. Throughput is modest, with no real-time requirement.

A4 — Scale — small (low-hundreds of portfolios/accounts).

We assume tens to low-hundreds of portfolios, a few thousand instruments, and daily volumes in the thousands of rows — comfortably single-node relational, needing no partitioning or scale-out.

A5 — Two sources today, source layer built to generalise.

We assume two providers now (Corner via Allocare, Julius Baer) but treat the provider/source dimension as first-class, so a third custodian can be added without schema change.

A6 — BJB PMS feed layout unknown; those tables are provisional.

We assume the BJB position/transaction CSV broadly mirrors Allocare's P73/P11 content, but its columns are not yet specified, so any BJB position or transaction table is provisional. The camt.053 (`.001.04`) and eDocs `.idx` structures are taken as known, since the BJB documentation specifies them.

A7 — Cash and securities arrive separately and must be reconciled.

We assume BJB cash arrives via camt.053 and securities via the PMS feed, while Allocare integrates both — so the same cash can appear in more than one source. Reconciliation is therefore a deliberate function, not an assumption that one source is canonical.

A8 — Read / consolidate / reconcile, not write-back.

We assume the database consumes export-direction (E) data for reporting and reconciliation and does *not* write back to Allocare or BJB.

A9 — Operational and security baseline.

We assume the firm must meet BJB's mandated SFTP algorithm set effective 28 June 2026, retains eDocs promptly given the 20-working-day purge, and operates under Swiss data-handling expectations for a Lugano wealth manager. These constrain the platform, not the data model.

2 Requirements

The following requirements follow from the assumptions above. Functional requirements describe what the system must do; non-functional requirements describe how it must behave.

2.1 Functional Requirements

Ingestion & Delivery

FR-1 Retrieve files from both sources over SFTP on a scheduled batch cycle aligned to BJB's pre-08:00 CET, T-1 window, handling BJB's Tuesday–Saturday calendar: weekend non-delivery is normal, a missing weekday file is an exception (flagged per FR-23).

FR-2 Parse all in-scope Allocare GDF file types (P01, P02, P03, P07, P11, P31, P32, P35, P36, P73), respecting the four-line `!` header, optional `#` comment lines, tab delimiting, and the leading edit-function field.

FR-3 Parse the BJB PMS positions/transactions CSV feed. *(Provisional — finalised on receipt of BJB's record layout; see assumption A6.)*

FR-4 Shred BJB camt.053 (`.001.04`) XML into its hierarchy: Group Header (A) → Statement and balances (B) → Entry (C) → Entry Details (D), accepting balance-only statements that contain no C or D level.

FR-5 Unpack BJB eDocs ZIP packages: extract the PDFs and parse the semicolon-delimited `.idx` index (`ClientNumber ; Date ; Layout ; DocumentType ; Account ; Filename`).

FR-6 Parse version-aware: read the declared file/message version (GDF header `Vx.x`; camt `.001.04`) and select the matching layout rather than assuming a fixed column map.

FR-7 Land every feed verbatim in a source-shaped staging area before transformation, retaining the original record and its file lineage.

Identity & Reference

FR-8 Maintain identity crosswalks: (a) mapping all instrument identifier systems (ISIN, Valor, WKN, CUSIP, SEDOL, Bloomberg, Ticker, `ASSET`, BJB instrument id, custom) to a single internal `instrument_key`, honouring the P01 `IdtfSys` field that declares the authoritative identifier per record; and (b) linking the Allocare portfolio name and `Client` to the BJB account number, `ClientNumber`, and IBAN.

FR-9 Maintain a code-value dictionary and harmonise the three code systems — Allocare numeric codes (asset category, transaction type, movement type, etc.), ISO 20022 bank transaction codes (`BkTxCd`), and eDocs letter codes (B/V/C/D/K/X/R/G) — into one internal taxonomy.

FR-10 Treat sentinel values (`Unknown`, `??+???+5`, `???`) as null-equivalent rather than as data.

Core Data Management

FR-11 Honour edit functions I/U/D, including full-replace semantics where the source requires it (e.g. P03/P46 delete-all-then-reload per investment).

FR-12 Store transactions in the GDF two-row model — a transaction header plus one or more ordered movement lines — preserving line order and the `Intern1`/`Intern2` links needed to reassemble them.

FR-13 Store effective-dated reference data as history rather than single values (e.g. the P32 portfolio reference-currency rows, including the blank-dated baseline).

FR-14 Store positions as dated snapshots (Allocare P73 and the BJB PMS positions), capturing the `Date Qualifier` (T/V) / booking-versus-value distinction.

FR-15 Store cash statements, entries, and opening/closing balances from camt.053.

FR-16 Maintain a document register from the eDocs index, with blob references to the archived PDFs.

FR-17 Tag every core record with its source/provider and retain its native keys.

FR-18 Preserve signed quantities and full decimal precision (prices to five decimal places, FX to eight) using decimal types, never floating point.

FR-19 Represent localised instrument names (EN/FR/IT/DE) with the documented fallback rule: a blank localised name is not missing data.

Reconciliation & Consolidation

FR-20 Reconcile cash and securities across overlapping sources (e.g. camt closing balance vs. Allocare cash account vs. P73 position value) under explicit precedence rules rather than assuming one canonical source, flagging discrepancies as exceptions for review.

FR-21 Produce a consolidated cross-bank view per client and portfolio.

FR-22 Support multi-currency consolidation using each portfolio's reference currency and the FX time series (P07).

Operational Functions

FR-23 Detect and report missing, late, or duplicate files against the expected delivery schedule.

FR-24 Be idempotent: re-processing the same file must not double-load data.

FR-25 Handle partial or failed loads gracefully — load valid records, isolate and report rejected ones (mirroring GDF's mark-bad-message-and-continue behaviour).

FR-26 Record load lineage for every file: name, type, version, source, creation timestamp, row counts, and load outcome.

2.2 Non-Functional Requirements

NFR-1 *(Performance / throughput)* Complete each daily batch load comfortably before the business day, sized for small scale (low-hundreds of portfolios, a few thousand instruments, thousands of daily rows) on a single-node relational database.

NFR-2 *(Processing model)* Batch, not real-time; no intraday or streaming requirement is assumed.

NFR-3 *(Security — transport)* Meet BJB's mandated SFTP algorithm set (key exchange / cipher / MAC) effective 28 June 2026, with SSH key-pair management and fixed-IP whitelisting.

NFR-4 *(Security — data)* Protect client and portfolio data at rest and in transit, consistent with Swiss data-protection expectations (FADP) and the confidentiality norms of a Lugano wealth manager.

NFR-5 *(Auditability)* Keep an immutable staging layer and full lineage so that any core record can be traced back to the exact source file and row.

NFR-6 *(Extensibility)* Add a new custody source or feed type without changing the core schema — the provider/source dimension is first-class (assumption A5).

NFR-7 *(Maintainability)* Drive parsers from configurable, version-aware layout definitions so that format or version changes are configuration, not code rewrites.

NFR-8 *(Data integrity)* Enforce referential integrity across the crosswalks and core, and the decimal-precision rule of FR-18, with no silent type coercion.

NFR-9 *(Reliability / recoverability)* Reprocess any day from staging without re-fetching, and tolerate expected gaps (weekend non-delivery) without raising false alarms.

NFR-10 *(Retention / archival)* Persist eDocs promptly and permanently in the firm's store, independent of BJB's 20-working-day purge, under a defined internal retention policy.

NFR-11 *(Time correctness)* Handle timezone semantics correctly — CET-based BJB cutoffs and the timezone-less `yyyymmdd` / `yyyymmdd:hhmmss` Allocare dates — normalising to one internal convention.

NFR-12 *(Platform portability)* Keep the model RDBMS-agnostic (MS SQL Server a reasonable default, since Allocare runs on it) so that the firm's platform choice can be slotted in later (assumption A3).

2.3 Traceability Notes

- **FR-3** is the single provisional requirement; it is finalised when the BJB CSV layout arrives (assumption A6).
- **FR-8** and **FR-9** are the crosswalk-and-harmonisation core; under Route A (assumption A1) they collapse to identity resolution within GDF only.
- **NFR-3** carries the only hard external deadline: 28 June 2026.